



Newington College

2004

TRIAL HSC EXAMINATION

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks: 120

- Attempt Questions 1–10
- All questions are of equal value

Question 1 (12 marks)

Marks

- (a) Find the value of

1

$$\log_e 3.5 - \frac{\pi}{\sqrt{e^3}},$$

correcting your answer to 3 significant figures.

2

- (b) Find the primitive of
- x^{-2}
- .

- (c) Solve
- $\frac{3t}{t-5} = \frac{2}{5}$
- .

2

- (d) Jai sold his car for \$5400, which was 18% less than its original cost.

2

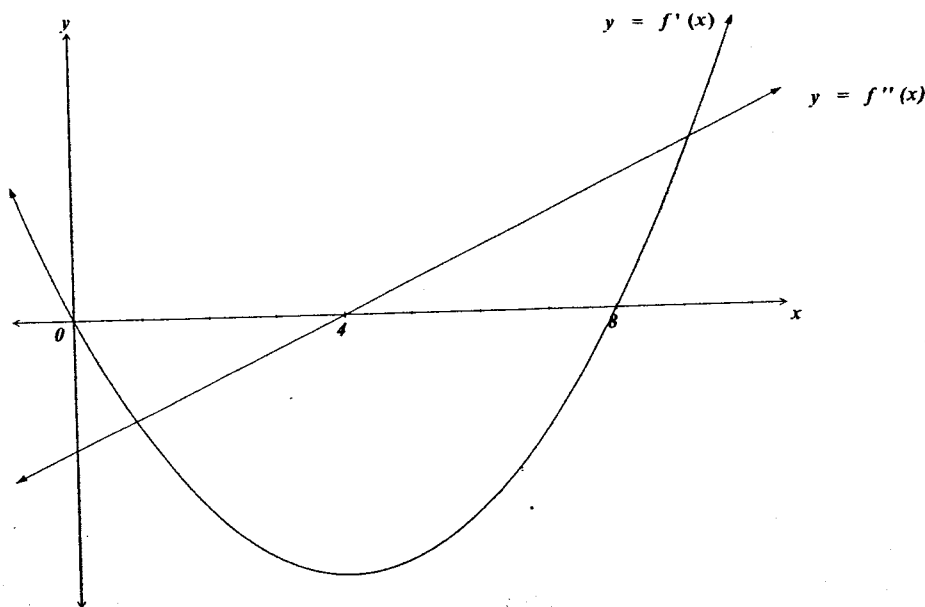
How much did the car originally cost?

- (e) Solve
- $|2 - 3x| \geq 1$
- .

3

- (f)

2



The graphs of the first and second derivatives of the curve $y = f(x)$ are shown in the diagram.

Write down the x coordinates of the stationary points and determine their nature.

Question 2 (12 marks) Use a SEPARATE writing booklet.

Marks

(a) Factorise $1 - 27a^3$.

1

(b) Differentiate the following with respect to x :

1

(i) $y = \sin^3 x$

1

(ii) $y = \log_e(3x + 1)$

1

(iii) $y = 2e^{-4x}$

(c) Find the value of p

1

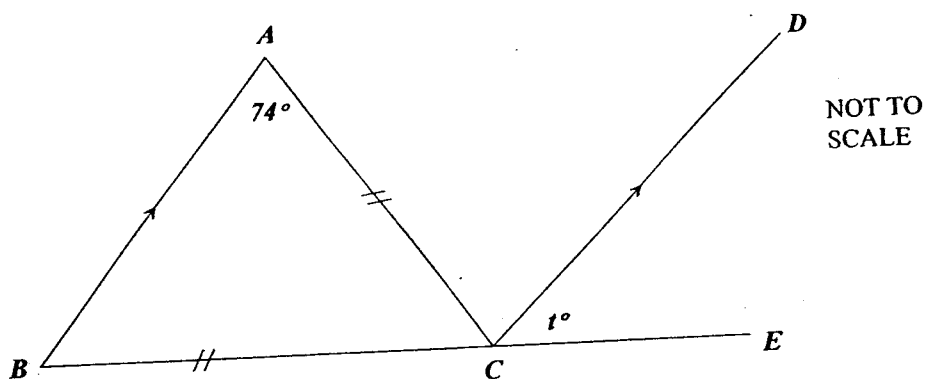
(i) $\log\left(\frac{1}{x^2}\right) = p \log x$

(ii) $6^p = 3$, correct your answer to 2 decimal places.

2

(d) Find the value of the pronumeral, giving reasons.

2



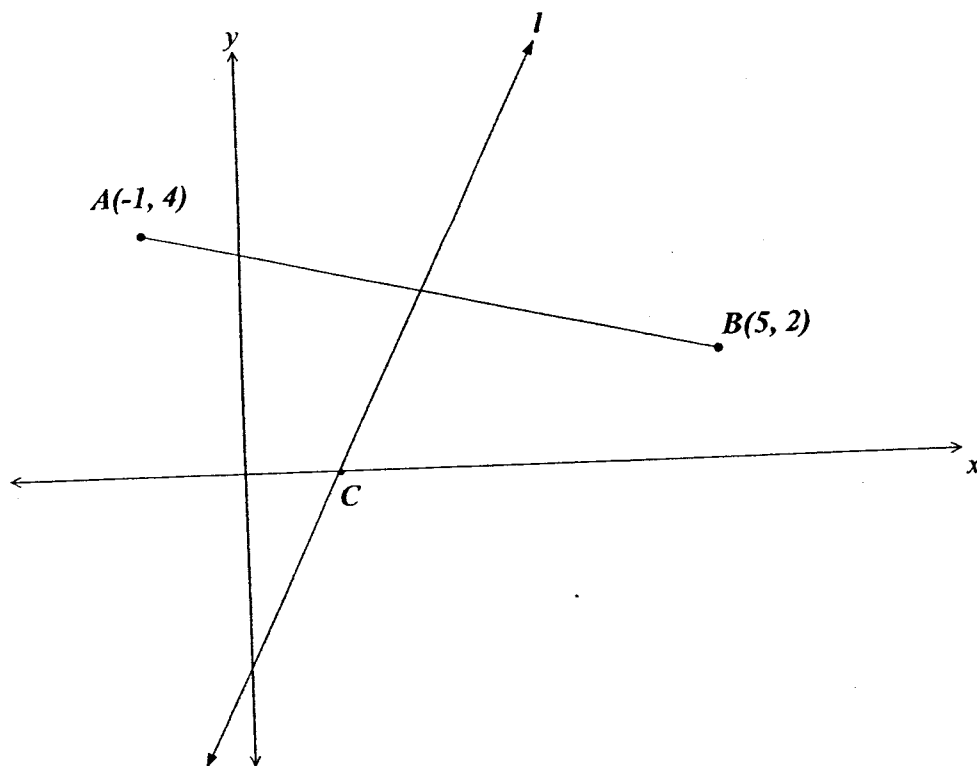
(e) Find all the real numbers x which satisfy the equation

3

$$x^4 = 4(x^2 + 8)$$

Question 3 (12 marks) Use a SEPARATE writing booklet. **Marks**

- (a) The diagram below shows the points $A(-1, 4)$ and $B(5, 2)$. The line l has the equation $3x - y - 3 = 0$ and cuts the x -axis at C .



- | | | |
|-------|--|---|
| (i) | Show that the length of AB is $2\sqrt{10}$ units. | 1 |
| (ii) | Find the co-ordinates of M , the midpoint of AB . | 1 |
| (iii) | Find the gradient of AB . | 1 |
| (iv) | Show that the equation of AB is $x + 3y - 11 = 0$. | 1 |
| (v) | Prove that l is the perpendicular bisector of AB . | 2 |
| (vi) | Find the co-ordinates of C . | 1 |
| (vii) | Write down the equation of the circle with AB as the diameter. | 1 |

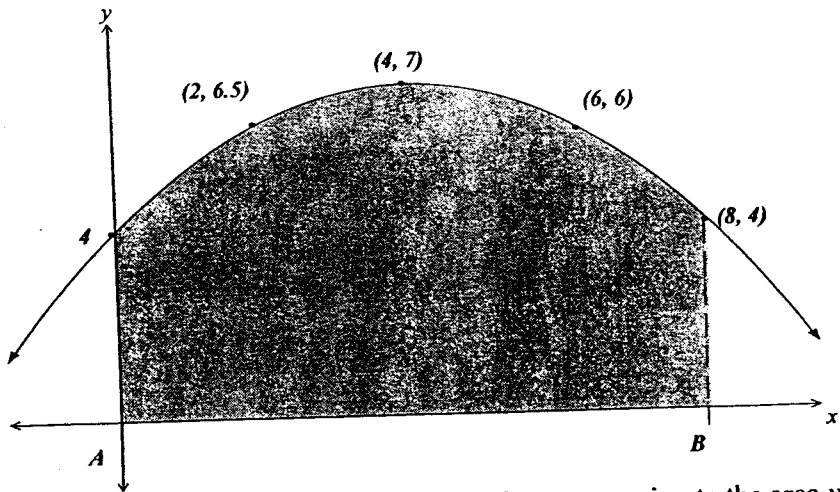
Question 3 continues on page 4

Question 3 (continued)

Marks

(b)

3



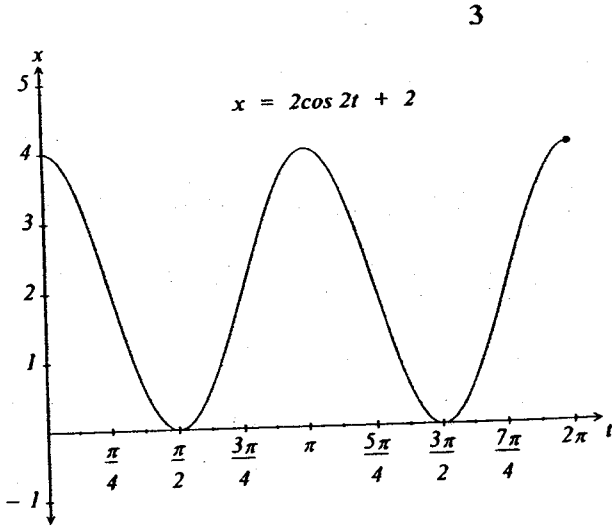
Using Simpson's Rule with 5 function values, approximate the area under the curve $y = f(x)$ from $x = A$ to $x = B$, as shown in the diagram above.

(c) Find $\int \frac{3x^2}{1+x^3} dx$.

1

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) A particle moves according to the displacement function $x = 2 \cos 2t + 2$ for $0 \leq t \leq 2\pi$, for x metres and t seconds, as shown at right:
- Find (i) when the particle is at rest.
(ii) when the particle is moving away from the origin.



Question 4 continues on page 5

Question 4 (continued)

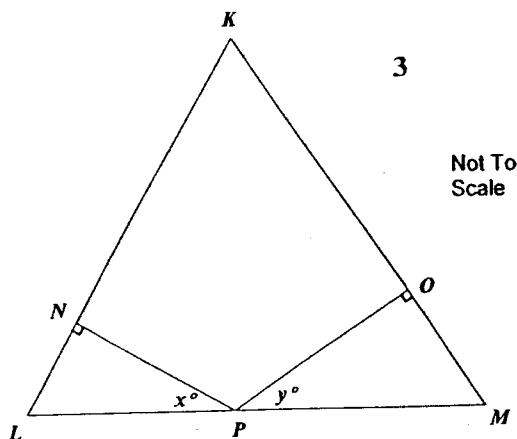
Marks

- b) A parabola, whose equation is of the form $y = Ax^2$ (where A is a constant) 3
has the line $20x - y + 20 = 0$ as a tangent.

Find the value of A .

- (c) Evaluate $\int_1^2 \frac{1}{3} e^{2.5x} dx$ 3

- (d) KLM is an isosceles triangle with
 $KL = KM$. The lines NP and OP are
constructed perpendicular to KL and KM
respectively, as shown, from the point P ,
the midpoint of LM .

Prove $x^\circ = y^\circ$.

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the function $f(x) = (x-4)(x+2)^2$
- (i) Show that $f'(x) = 3x^2 - 12$. 2
- (ii) Find the co-ordinates of the stationary points of the curve $y = f(x)$ 2
and determine their nature.
- (iii) Find any point(s) of inflexion. 1
- (iv) Sketch the graph of the curve $y = f(x)$ showing all intercepts, 2
stationary points and points of inflexion.
- (v) Find all the values of x for which the graph $y = f(x)$ 2
is concave up.

Question 5 continues on page 6

Question 5 (continued)

Marks

- (b) The first two terms of a geometric sequence are $\sqrt{3}-1$ and $\sqrt{3}+1$. 3

Find the common ratio and the fourth term of this sequence in simplest surd form.

Discuss whether the geometric sequence has a limiting sum.

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) What is the domain and range of $y = \sqrt{5-2x}$? 1

- (b) Find the value of C if the equation $4x^2 - 12x + C = 0$ has equal roots. 2

- (c) If α and β are the roots of the quadratic equation $x^2 - 5x - 4 = 0$,

find the values of :

(i) $\alpha + \beta$. 1

(ii) $\alpha\beta$. 1

(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$. 2

(iv) $\alpha^2 + \beta^2$. 2

- (d) For the parabola $x^2 = 8(y-3)$, find the

(i) co-ordinates of the vertex. 1

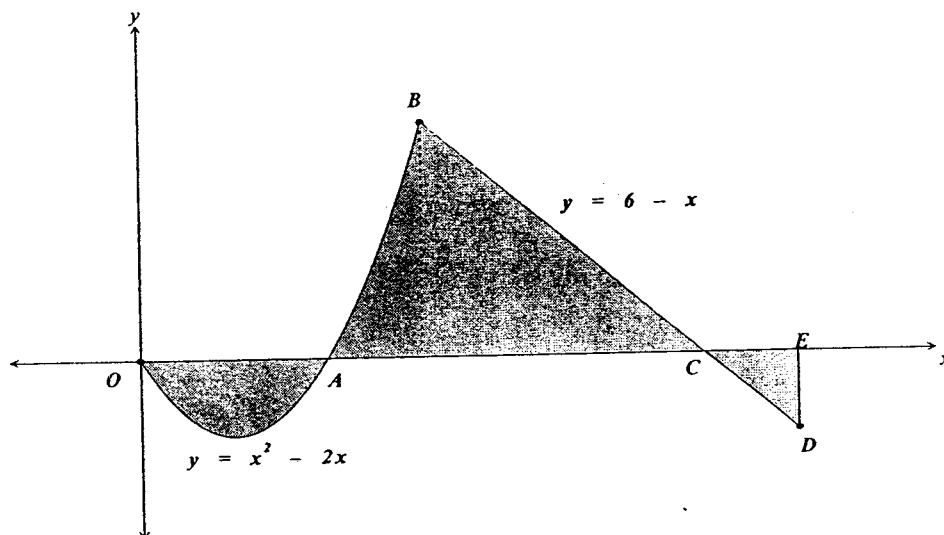
(ii) focal length. 1

(iii) equation of the directrix. 1

Question 7 (12 marks) Use a SEPARATE writing booklet. **Marks**

- (a) The shaded region $OABCDE$ is bounded by the lines $x = 0$ and $x = 7$, the curve $y = x^2 - 2x$, the line $y = 6 - x$ and the x -axis.

- (i) Find the co-ordinates of the points A , C and D . 3
- (ii) Show that the co-ordinates of B are $(3, 3)$. 2
- (iii) Calculate the area of the shaded region $OABCDE$. 3



- (b) If $f(x) = x^2$ then $f(x+3) \neq f(x) + f(3)$. 1

Find a function $g(x)$, such that $g(x+3) = g(x) + g(3)$.

- (c) A person saves \$10000 by investing \$100 at the beginning of each month at 6% p.a., compounded monthly.

- (i) Show that the expression for the amount saved by the end of the first three months is given by $A_3 = \$100(1.005^3 + 1.005^2 + 1.005)$. 1
- (ii) Hence, or otherwise, find the number of months required to save \$10000. 2

Question 8 (12 marks) Use a SEPARATE writing booklet.

Marks

- (a) Differentiate $\log_e(\cos^3 x)$ with respect to x . 3
- (b) If $\cos \theta = \frac{8}{17}$ and $\sin \theta < 0$, find the exact values for
- (i) $\sin \theta$ 2
- (ii) $\cot \theta$ 1
- (c) (i) Find the point of intersection of the two curves $y = x^2$ and $y = \frac{8}{x}$. 1
- (ii) Find the gradients of the tangents to both the curves at this point. 2
- (iii) Find the angles of inclination of both tangents to the positive x -axis 2
(Give your answer to the nearest degree).
- (iv) Hence find the acute angle between the two tangents. 1

Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) (i) On the same set of axes sketch $y = |2x - 3|$ and $y = -x$. 2
- (ii) Hence, or otherwise, solve $|2x - 3| < -x$. 1

Question 9 continues on page 9

(c)

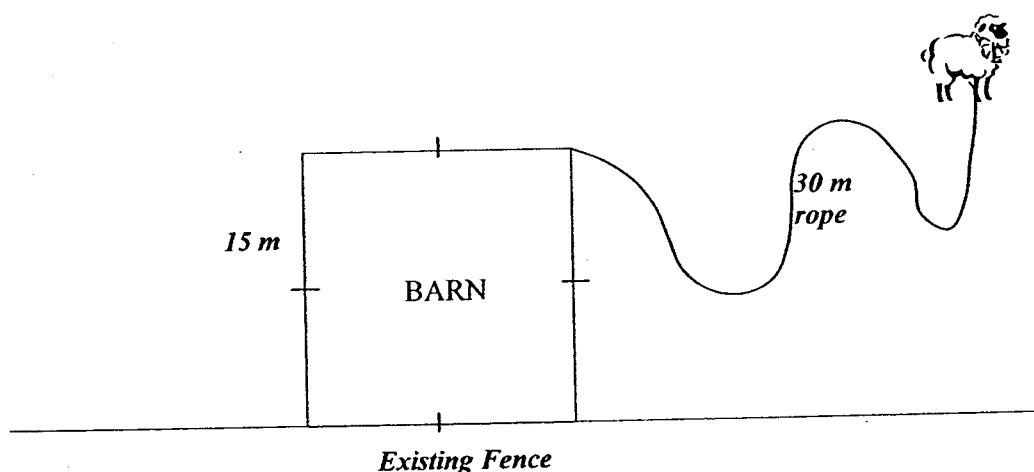
Question 9 (continued)

Marks

(b) A sheep is tied to the side of a square shaped barn as shown in the diagram.

The rope is 30 m long and the side length of the barn is 15 m. On one side of the barn is a fence line as shown.

- (i) Draw a diagram that shows the entire area over which the sheep can graze. (Not including the Barn) 2
- (ii) If the sheep can graze over the entire area but not inside the barn find the maximum grazing area available for the sheep. 4



- (c) The population of rats in a barn increases exponentially according to the formula $P = P_0 e^{kt}$, where P_0 is the original population and t is the time in months and k is a constant. 3

If the initial population of 20 increases to 100 in 3 months, how long will it take to reach a population in excess of 2000 at this rate?

Question 10 (12 marks) Use a SEPARATE writing booklet. **Marks**

(a) The probability of tossing a 5 on a biased six-sided die is $\frac{2}{7}$. The die is

tossed 3 times. Find the probability of tossing:

(i) one 5.

1

(ii) no 5's.

1

(iii) at least one 5.

1

The biased die is tossed n times.

(iv) Write an expression for the probability of tossing at least one 5.

1

(v) Find the value of n if the probability of tossing at least one 5 is

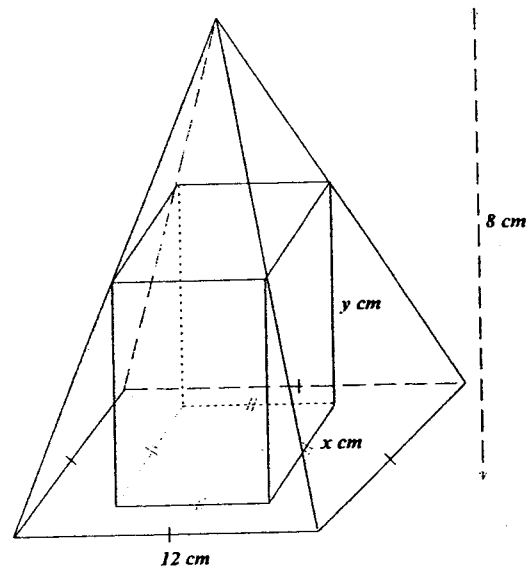
2

greater than 99%.

Question 10 continues on page 11

Question 10 (continued)

- (b) A rectangular prism is constructed within a square-based triangular pyramid as shown in the diagram:



- (i) Using similar triangles, or otherwise, show that 2
- $$y = 8 - \frac{2x}{3}.$$
- (ii) Let the volume of the rectangular prism be $V \text{ cm}^3$, then show that 2
- $$\frac{dV}{dx} = 2x(8 - x).$$
- (iii) Hence, find the maximum volume of the rectangular prism. 2

End of paper

Question 1 (12 marks)	Marks
(a) $\log_e 3.5 - \frac{\pi}{\sqrt{e^3}} = 0.552$ (correct to 3 significant figures)	1
(b) $\int x^{-2} dx = (-1)x^{-1} + c$ $= -\frac{1}{x} + c$	2
(c) $\frac{3t}{t-5} = \frac{2}{5}$ $15t = 2(t-5)$ $15t = 2t - 10$ $13t = -10$ $t = -\frac{10}{13}$	2
(d) 82% of original cost = \$5400 1% of original cost = \$5400/82 100% of original cost = \$5400 $\times$ 100/82 = \$6585.37 (to nearest cent)	2
(e) $ 2-3x \geq 1$ $2-3x \geq 1$ or $-(2-3x) \geq 1$ $3x \leq 1$ or $-2+3x \geq 1$ $x \leq \frac{1}{3}$ or $3x \geq 3$ $x \leq \frac{1}{3}$ or $x \geq 1$	3
(f) $f'(x) = 0$ at $x = 0$ and $x = 8$, hence these are stationary points. At $x = 0$, $f''(x) < 0$, hence this is a maximum turning point, and at $x = 8$, $f''(x) > 0$, so this is a minimum turning point.	2

(d)

Question 2 (12 marks) Start this question on a new page

Marks

(a) $1 - 27a^3 = (1 - 3a)(1 + 3a + 9a^2)$.

1

(b) (i)

$$y = \sin^3 x$$

$$\frac{dy}{dx} = \cos x \times 3\sin^2 x$$

1

$$\frac{dy}{dx} = 3\cos x \sin^2 x$$

(ii)

$$y = \log_e(3x + 1)$$

$$\frac{dy}{dx} = 3 \times \frac{1}{3x + 1}$$

1

$$\frac{dy}{dx} = \frac{3}{3x + 1}$$

(e)

(iii)

$$y = 2e^{-4x}$$

$$\frac{dy}{dx} = 2 \times (-4)e^{-4x}$$

1

$$\frac{dy}{dx} = -8e^{-4x}$$

(c)

(i)

$$\log\left(\frac{1}{x^2}\right) = p \log x$$

1

$$\frac{1}{x^2} = x^p$$

$$x^{-2} = x^p$$

$$p = -2$$

(ii)

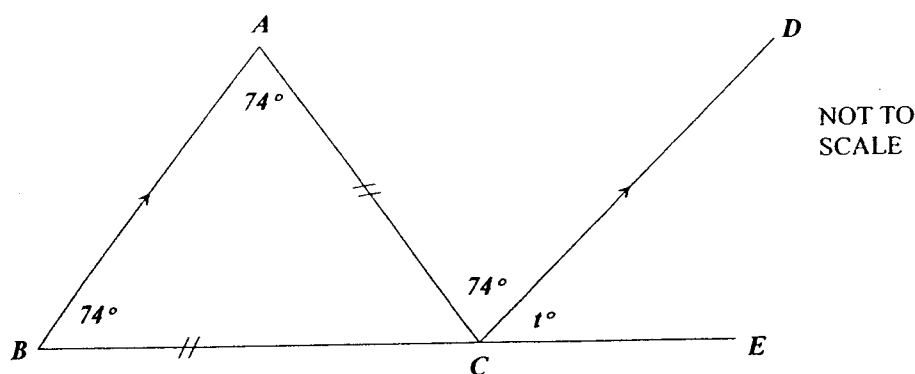
$$6^p = 3$$

$$p = \frac{\log 3}{\log 6}$$

2

$$p = 0.61 \text{ (correct to 2 decimal places)}$$

(d)



2

$$\angle ABC = \angle BAC = 74^\circ \text{ (equal angles opp equal sides)}$$

$$\angle BAC = \angle ACD = 74^\circ \text{ (alt angles equal, } BA \parallel CD \text{)}$$

$$\angle DCE = \angle BAC + \angle ABC - \angle ACD \text{ (exterior angle theorem)}$$

$$t = 74$$

(e)

$$x^4 = 4(x^2 + 8)$$

$$x^4 - 4x^2 - 32 = 0$$

$$(x^2 - 8)(x^2 + 4) = 0$$

$$(x - \sqrt{8})(x + \sqrt{8})(x^2 + 4) = 0$$

$$x = \pm\sqrt{8}$$

3

Question 3 (12 marks)

(a)

(i)

$$d_{AB} = \sqrt{(5+1)^2 + (2-4)^2}$$

$$d_{AB} = \sqrt{36+4}$$

$$d_{AB} = 2\sqrt{10}$$

1

$$(ii) \quad M = \left(\frac{-1+5}{2}, \frac{4+2}{2} \right) = (2, 3)$$

1

$$(iii) \quad m_{AB} = \frac{4-2}{-1-5} = \frac{2}{-6} = -\frac{1}{3}$$

1

Question 3 (12 marks)

(a) (iv)

$$y - y_1 = m_{AB}(x - x_1)$$

1

$$y - 2 = -\frac{1}{3}(x - 5)$$

$$3y - 6 = -x + 5$$

$$x + 3y - 11 = 0$$

(v)

At $M(2, 3)$, for $l: 3x - y - 3 = 0$

$$LHS = 3(2) - 3 - 3 = 0$$

2

$$RHS = 0$$

So, M lies on l .

$$m_l = 3$$

$$m_{AB} \times m_l = 3 \times \left(-\frac{1}{3}\right) = -1$$

So, l is perpendicular bisector.(vi) At C , let $y = 0$,

$$3x - y - 3 = 0$$

1

$$3x - 3 = 0$$

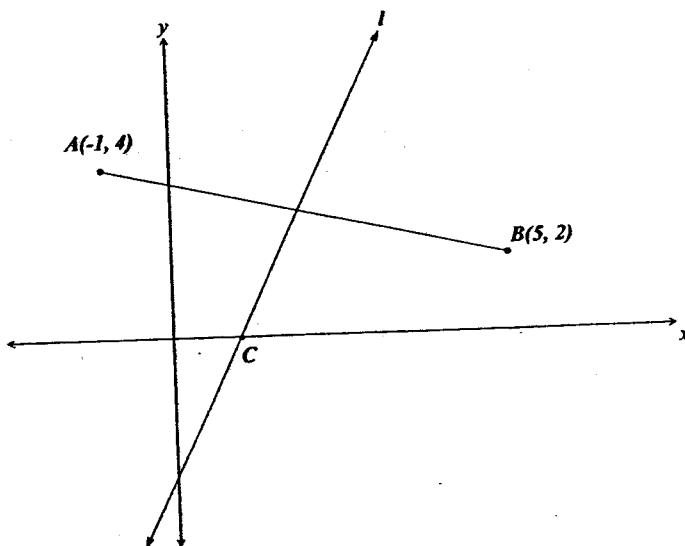
$$x = 1$$

(vii) If AB is the diameter then M is the centre i.e. $(2, 3)$ and radius is

$$\frac{1}{2}AB = \frac{1}{2}(2\sqrt{10}) = \sqrt{10}, \text{ so equation of circle is}$$

1

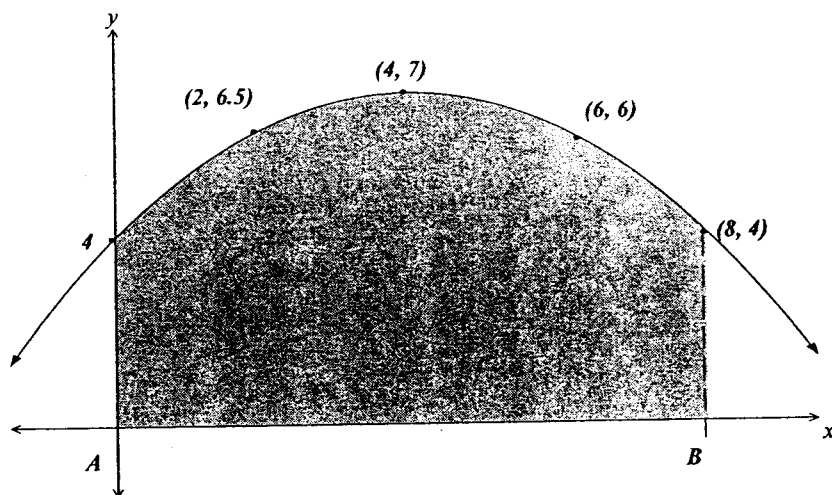
$$(x - 2)^2 + (y - 3)^2 = 10$$



Q3 cont.../page 4

Question 3 (cont.)

b)



x	0	2	4	6	8
$f(x)$	4	6.5	7	6	4

$$\text{Area} \cong \frac{h}{3} \{y_0 + 4y_1 + 2y_2 + 4y_3 + y_4\}, \text{ where } h = 2$$

$$\cong \frac{2}{3} \{4 + 4 \times 6.5 + 2 \times 7 + 4 \times 6 + 4\}$$

$$\cong \frac{2}{3} \{4 + 26 + 14 + 24 + 4\}$$

$$\cong 48 \text{ units}^2$$

3

$$(c) \int \frac{3x^2}{1+x^3} dx = \log_e(1+x^3) + c.$$

1

Q4... /page6

Question 4 (12 marks) Start this question on a new page

(a)

(i) At rest, if $v = 0$,

$$x = 2 \cos 2t + 2$$

$$\frac{dx}{dt} = v$$

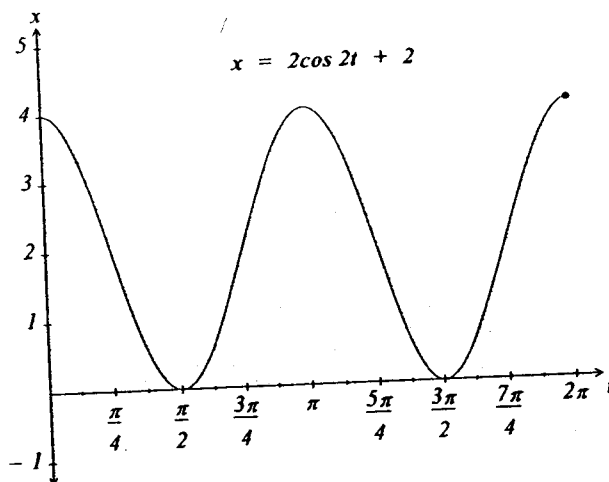
$$v = -4 \sin 2t$$

$$-4 \sin 2t = 0$$

$$\sin 2t = 0$$

$$2t = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$



3

(ii)

From the graph, if

displacement increases then the object is moving away from the origin,

$$\text{i.e. } \frac{\pi}{2} \leq t \leq \pi, \frac{3\pi}{2} \leq t \leq 2\pi.$$

(b)

$$y = Ax^2$$

$$y = 20x + 20$$

$$Ax^2 - 20x - 20 = 0, \text{ has only one solution, i.e. } b^2 - 4ac = 0$$

3

$$(-20)^2 + 80A = 0$$

$$80A = -400$$

$$A = -5$$

(ii)

(c)

$$\int_1^2 \frac{1}{3} e^{2.5x} dx = \frac{1}{3} \left[\frac{e^{2.5x}}{2.5} \right]_1^2$$

$$= \frac{1}{3} \left[\frac{e^5 - e^{2.5}}{2.5} \right]$$

$$= 18.16 \text{ (correct to 2 decimal places)}$$

3

(iii)

Q4 cont.../page 7

Question 4 (cont.)

Marks

(d)

$$\angle NLP = \angle OMP \text{ (equal angles in isosceles triangle)}$$

$$\angle NLP = 90^\circ - x^\circ \text{ (complementary angle)}$$

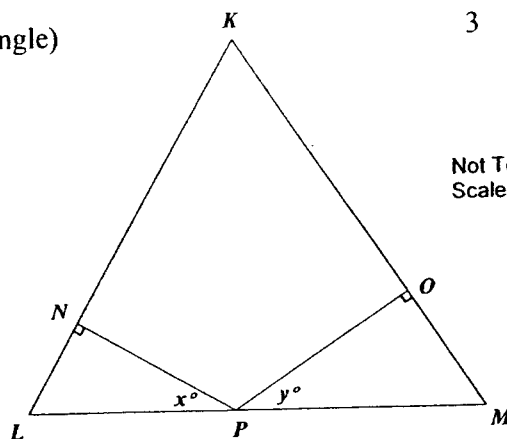
$$\angle OMP = 90^\circ - y^\circ \text{ (complementary angle)}$$

$$90^\circ - x^\circ = 90^\circ - y^\circ$$

$$x^\circ = y^\circ$$

3

Not To Scale



Question 5

(a) (i)

$$f(x) = (x-4)(x+2)^2$$

2

$$f(x) = (x-4)(x^2 + 4x + 4)$$

$$f(x) = x^3 + 4x^2 + 4x - 4x^2 - 16x - 16$$

$$f(x) = x^3 - 12x - 16$$

$$f'(x) = 3x^2 - 12$$

$$f''(x) = 6x$$

(ii)

$$\text{Let } f'(x) = 0$$

2

$$3x^2 = 12$$

$$x^2 = 4$$

$$x = \pm 2$$

At $x = 2$, $f''(x) = 12 > 0$ so min. turning point.

At $x = -2$, $f''(x) = -12 < 0$ so max. turning point.

(iii)

$$\text{If } f''(x) = 0 \text{ then}$$

$$6x = 0$$

$$x = 0$$

$$y = (-4)(2)^2$$

$$y = -16$$

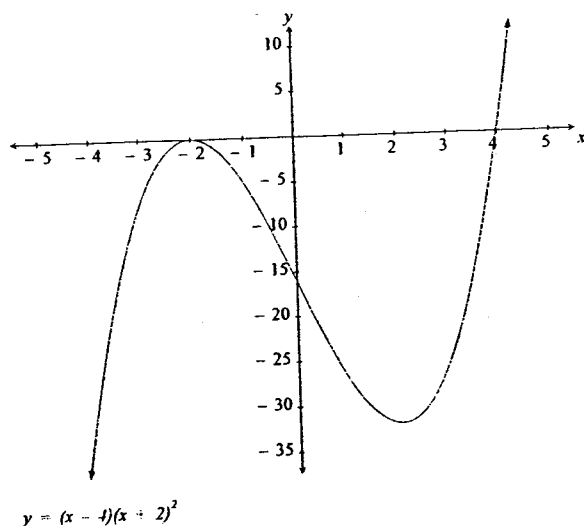
Point of inflexion at $(0, -16)$

1

Q5 cont.../page 5

Question 5 (cont.)

(a) (iv)



2

(v) If $f(x)$ is concave up then $f''(x) < 0$, hence $x > 0$.

2

(b) Common ratio $= \frac{\sqrt{3}+1}{\sqrt{3}-1}$

$$\begin{aligned} r &= \frac{\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1} \\ &= \frac{3+2\sqrt{3}+1}{2} \\ &= 2+\sqrt{3} \end{aligned}$$

3

$$T_4 = ar^{n-1}$$

$$\begin{aligned} &= (\sqrt{3}-1)(2+\sqrt{3})^3 \\ &= 11\sqrt{3}+19 \end{aligned}$$

As $|r| = 2+\sqrt{3} > 1$ then no limiting sum.

Question 6

1

(a) Domain:

Range:

$$5-2x \geq 0$$

$$2x \leq 5$$

$$x \leq 2\frac{1}{2}$$

$$y \geq 0$$

Q6 cont.../page 6

Question 6 (cont.)

(b) For $4x^2 - 12x + C = 0$ to have equal roots,

2

$$\Delta = b^2 - 4ac = 0$$

$$144 - 16C = 0$$

$$16C = 144$$

$$C = 9$$

(c) For $x^2 - 5x - 4 = 0$,

$$(i) \quad \alpha + \beta = -\frac{b}{a} = -\frac{(-5)}{1} = 5.$$

1

$$(ii) \quad \alpha\beta = \frac{c}{a} = \frac{(-4)}{1} = -4.$$

1

$$(iii) \quad \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{5}{(-4)} = -\frac{5}{4}.$$

2

$$(iv) \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = (5)^2 - 2(-4) = 33.$$

2

(d) If $(x-h)^2 = 4a(y-k)$ then for the parabola $x^2 = 8(y-3)$,

$$(i) \quad \text{Vertex} = (h, k) = (0, 3)$$

1

$$(ii) \quad \text{Focal length} = a = 2.$$

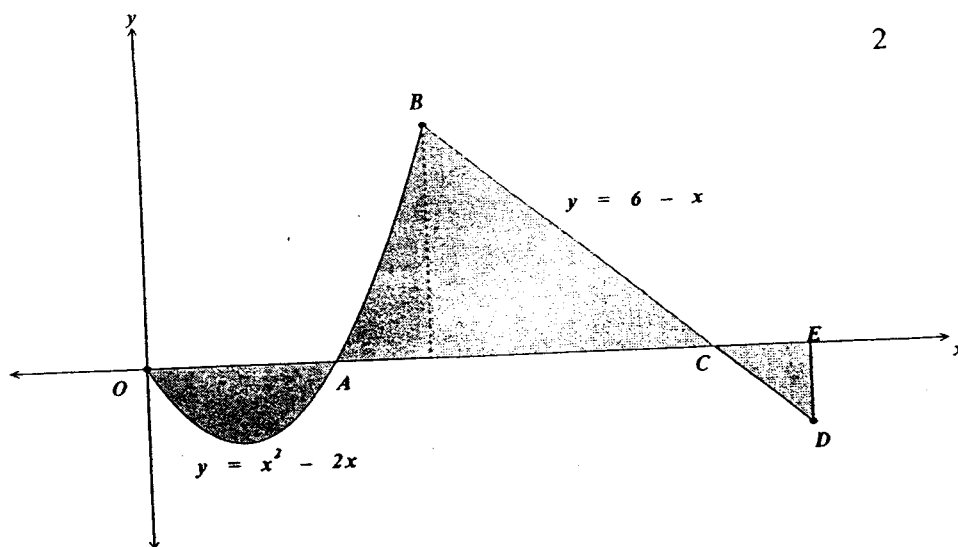
1

$$(iii) \quad \text{Equation of the directrix: } y = k - a = 1$$

1

Q7.../page 10

Question 7



- (a) (i) At A,
 $x^2 - 2x = 0$
 $x(x - 2) = 0$, So, A (2, 0)
 $x = 0, 2$
- At C,
 $6 - x = 0$, So C (6, 0)
 $x = 6$
- At D, $x = 7$
 $y = 6 - 7 = -1$, So, D (7, -1)
- (ii) At B,
 $x^2 - 2x = 6 - x$
 $x^2 - x - 6 = 0$
 $(x - 3)(x + 2) = 0$, So, B (3, 3)
 $x = 3, -2$
 If $x = 3$, then $y = 3$

Q7 cont.../page11

Question 7 (cont.)

(iii)

$$\begin{aligned}
 \text{Area} &= \left| \int_0^2 x^2 - 2x \, dx \right| + \int_2^3 x^2 - 2x \, dx + \frac{1}{2} \times (6-3) \times 3 + \frac{1}{2} \times (7-6) \times 1 \\
 &= \left| \left[\frac{x^3}{3} - x^2 \right]_0^2 \right| + \left[\frac{x^3}{3} - x^2 \right]_2^3 + \frac{9}{2} + \frac{1}{2} \\
 &= \left| \left(\frac{8}{3} - 4 \right) - 0 \right| + \left[\left(\frac{27}{3} - 9 \right) - \left(\frac{8}{3} - 4 \right) \right] + 5 \\
 &= \left| -\frac{4}{3} \right| + \frac{4}{3} + 5 \\
 &= 7\frac{2}{3} \text{ square units}
 \end{aligned}$$

3

(b) Typically, $f(x) = kx$, is one possible solution.

1

(c) (i)

$$\begin{aligned}
 A_1 &= 100(1.005) \\
 A_2 &= (A_1 + 100)(1.005) \\
 &= 100(1.005)^2 + 100(1.005) \\
 A_3 &= (A_2 + 100)(1.005) \\
 &= (\$100(1.005)^2 + \$100(1.005) + \$100)(1.005) \\
 &= \$100(1.005)^3 + \$100(1.005)^2 + \$100(1.005) \\
 &= \$100(1.005^3 + 1.005^2 + 1.005)
 \end{aligned}$$

1

(ii) If $A_n = \$10000$, then

$$\begin{aligned}
 A_n &= \$100(1.005^n + 1.005^{n-1} + \dots + 1.005) \\
 &= \frac{\$100(1.005)(1.005^n - 1)}{0.005}
 \end{aligned}$$

$$\frac{\$100(1.005)(1.005^n - 1)}{0.005} = 10000$$

$$1.005^n = \frac{\$10000 \times 0.005}{\$100(1.005)} + 1$$

$$n = \frac{\log_e \left[\frac{\$10000 \times 0.005}{\$100(1.005)} + 1 \right]}{\log_e (1.005)}$$

$$n = 80.96$$

$$n = 81 \text{ months}$$

2

Q8.../page 12

Question 8

(a)

$$\begin{aligned}\frac{d(\log_e(\cos^3 x))}{dx} &= -\sin x \times 3 \cos^2 x \times \frac{1}{\cos^3 x} \\ &= -3 \frac{\sin x}{\cos x} \\ &= -\tan x\end{aligned}$$

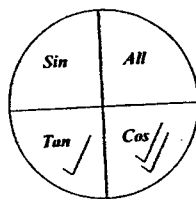
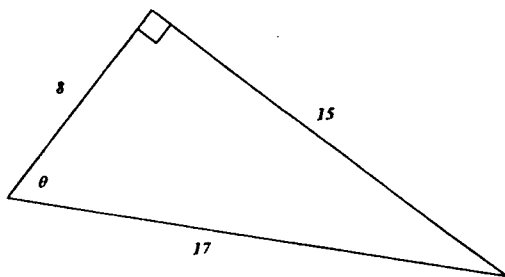
3

- (b) If $\cos \theta = \frac{8}{17}$ and $\sin \theta < 0$, then $\frac{3\pi}{2} \leq \theta \leq 2\pi$ (see diag)

Using Pythagoras' theorem,

$$x^2 = 17^2 - 8^2$$

$$x = 15$$



(i) $\sin \theta = -\frac{15}{17}$

2

(ii) $\cot \theta = -\frac{15}{8}$

1

- (c) (i) For the curves, $y = x^2$ and $y = \frac{8}{x}$,

$$x^2 = \frac{8}{x}$$

$$x^3 = 8$$

$$x = 2$$

$$y = 4$$

So, (2, 4) is the point of intersection.

1

(ii) For $y = \frac{8}{x}$, $\frac{dy}{dx} = -\frac{8}{x^2} = -\frac{8}{4} = -2 \dots\dots [m_1]$

2

For $y = x^2$, $\frac{dy}{dx} = 2x = 4 \dots\dots\dots [m_2]$

Q8 cont.../page 13

Q8 (cont.)

(c) (iii)

$$\tan \theta = 4$$

$$\theta = 76^\circ \text{ (to nearest degree)}$$

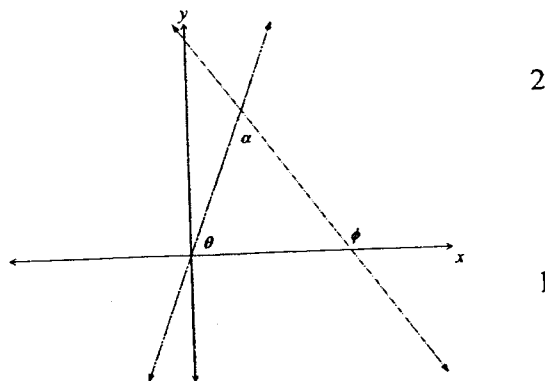
$$\tan \phi = -2$$

$$\phi = 117^\circ \text{ (to nearest degree)}$$

(iv)

$$\alpha = \phi - \theta \text{ (see diagram)}$$

$$\alpha = 41^\circ$$



Question 9

(a)

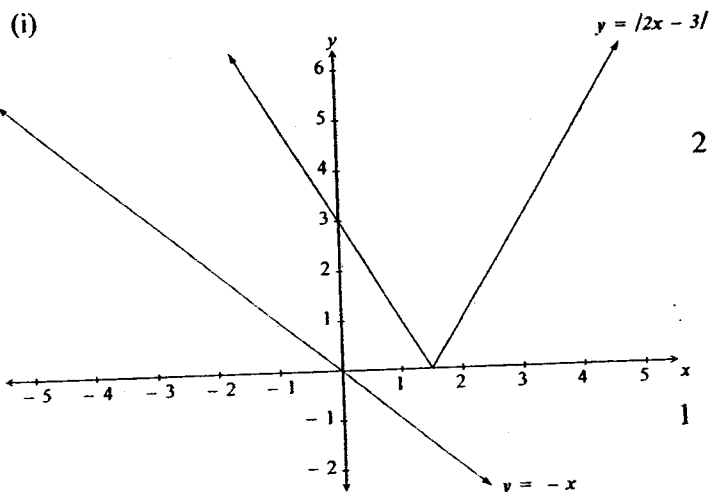
(ii)

From the graph it can be seen

that

$$|2x + 3| < -x$$

has no solutions.



(b)

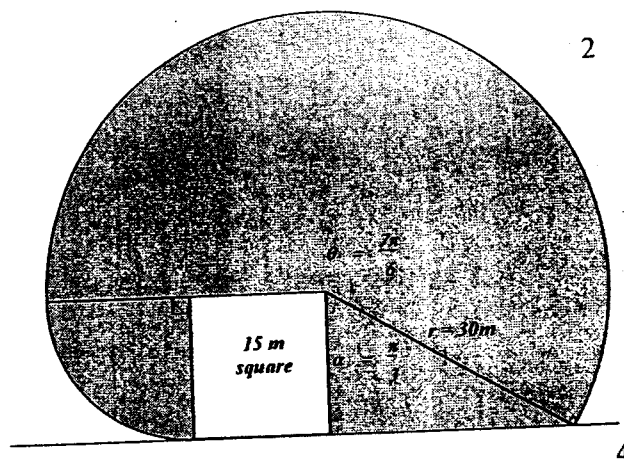
(ii)

Total Area =

$$\frac{\frac{7\pi}{6} \times 30^2}{2} + \frac{15^2 \times \pi}{4} + \frac{1}{2} \times 30 \times 15 \times \sin \frac{\pi}{3}$$

$$= 581\frac{1}{4} + 225\sqrt{3} \text{ sq. m.}$$

(i)



Q9 cont.../page 14

Q9 (cont.)

- (c) $P = P_0 e^{kt}$, if $P_0 = 20$, and $P = 100$ when $t = 3$.

$$100 = 20e^{3k}$$

$$e^{3k} = 5$$

$$k = \frac{\ln 5}{3}$$

3

Thus, if $P = 2000$ then

$$20e^{kt} = 2000$$

$$e^{kt} = 100$$

$$kt = \ln 100$$

$$t = \frac{\ln 100}{k}$$

$$t = \frac{3 \ln 100}{\ln 5}$$

$$t \approx 9 \text{ months}$$

Question 9

(v)

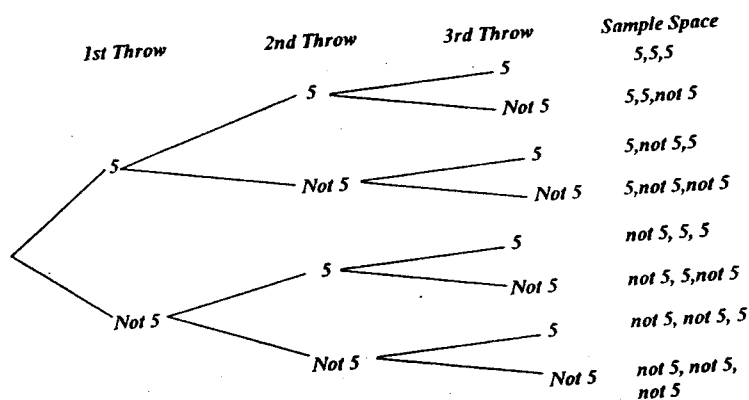
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(i)

Question 10

(a)



- (i) $P(\text{one } 5) = \frac{2}{7} \times \frac{5}{7} \times \frac{5}{7} + \frac{5}{7} \times \frac{2}{7} \times \frac{5}{7} + \frac{5}{7} \times \frac{5}{7} \times \frac{2}{7} = \frac{150}{343}$
- (ii) $P(\text{no } 5\text{'s}) = \frac{5}{7} \times \frac{5}{7} \times \frac{5}{7} = \frac{125}{343}$
- (iii) $P(\text{at least one } 5) = 1 - P(\text{no } 5\text{'s}) = 1 - \frac{125}{343} = \frac{218}{343}$
- (iv) $P(\text{at least one } 5) = 1 - \left(\frac{5}{7}\right)^n$

1

1

1

1

Q10 cont.../page 15

Question 9 (cont.)

- (v) If $P(\text{at least one } 5) > 0.99$ then

$$1 - \left(\frac{5}{7}\right)^n > 0.99$$

$$\left(\frac{5}{7}\right)^n < 0.01$$

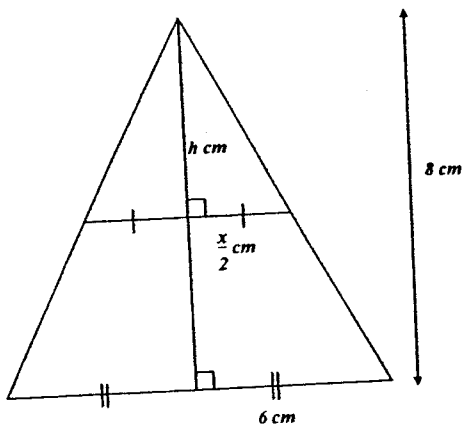
2

$$n = \frac{\log 0.01}{\log \left(\frac{5}{7}\right)}$$

$$n \approx 14 \text{ (nearest whole number)}$$

- (b) A rectangular prism is constructed within a square-based triangular pyramid as shown in the diagram:

- (i) By taking a vertical section through the prism:

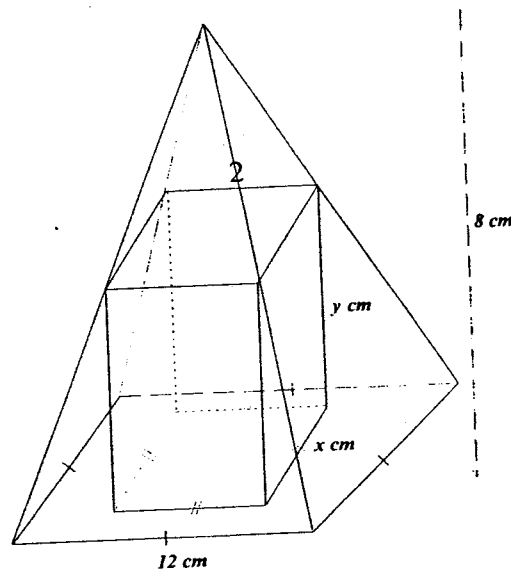


$$\frac{h}{8} = \frac{\frac{x}{2}}{6}$$

$$h = \frac{2x}{3}$$

$$y = 8 - h$$

$$y = 8 - \frac{2x}{3}$$



2

Q10 cont.../page 16

Q10 (cont.)

(b)

(ii)

$$V = x^2 y$$

$$V = x^2 \left(8 - \frac{2x}{3} \right)$$

2

$$V = 8x^2 - \frac{2x^3}{3}$$

$$\frac{dV}{dx} = 16x - 2x^2$$

$$\frac{dV}{dx} = 2x(8 - x)$$

(iii) If

$$\frac{dV}{dx} = 0$$

$$2x(8 - x) = 0$$

$$x = 0 \text{ or } 8$$

$$\frac{d^2V}{dx^2} = 16 - 4x$$

$$\text{If } x = 8 \text{ then } \frac{d^2V}{dx^2} < 0$$

$$\text{Maximum volume} = 64 \left(8 - \frac{2 \times 8}{3} \right)$$

$$= 170 \frac{2}{3} \text{ cm}^3$$

END OF EXAMINATION

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